

Name:

Collaborators:

Instructions: Worksheets are graded mostly on completion and partially on correctness. Please write complete solutions showing explanations and key steps to the following problems, unless it says otherwise.

Analyze Bifurcations in One Dimension

1. Fish Harvesting

An engineer at a fish hatchery has been using the logistic growth model for predicting the number of fish that a hatchery can expect to find in their pond.

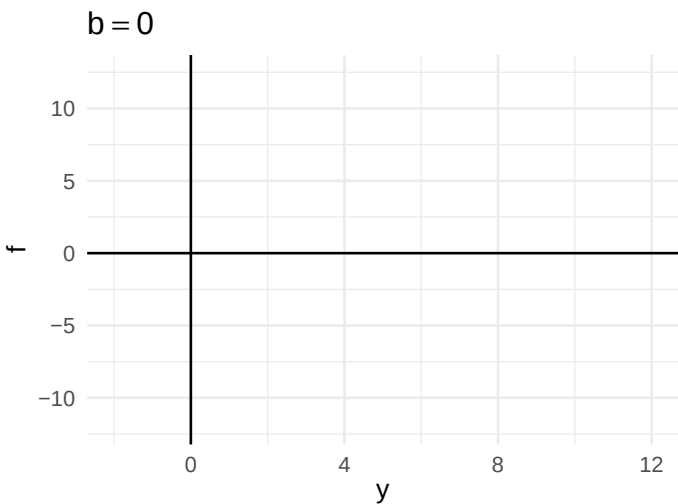
Let $y(t)$ denote the fish population (in thousands) at time t , with initial population $y(0) = y_0$. The population dynamics are modeled by the logistic growth ODE,

$$y' = 5y \left(1 - \frac{y}{10}\right) - b,$$

where the term $5y$ represents natural growth, $\left(1 - \frac{y}{10}\right)$ accounts for the environmental resistance with carrying capacity 10 (thousand), and the term $-b$ is a parameter with $b \geq 0$ is the fishing rate (thousand fish per unit time).

The goal here is to study how the solutions of the ODE change as the fishing rate b changes, and then interpret what this means in context.

- a. Use the blank axis shown below to create the phase diagram for $b = 0$ (no fishing). Then, label the stability of equilibriums.
- b. Define $f(y)$ and let $b = 0$. Then, determine the equilibrium solutions.



- c. Describe the behavior of solutions and what this means in context.

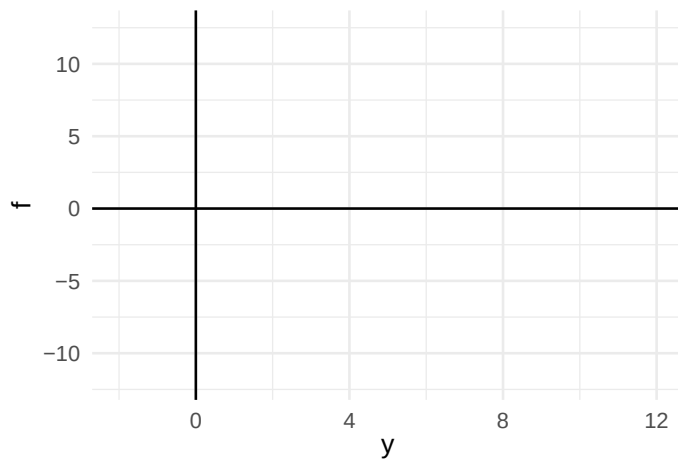
2. Bifurcation

Continued from Problem (1). Now, we analyze the solutions as we vary the parameter b of the given ODE.

- A *bifurcation* is a change in number of equilibriums or change in equilibrium stability as a parameter changes.
- A *bifurcation point* is the value of the parameter on which the change occurs.

a. Create a complete phase diagram of the given ODE for each designated b value.

$b = 3.125$



$b = 6.25$



$b = 12.5$



b. Define $f(y)$ and assume $b \geq 0$. Then, determine the equilibrium solutions. These equilibrium solutions should have the b term. Describe your observations as b increases.

c. Determine the bifurcation point. Then, describe what happens to the equilibriums at and beyond this point.

d. In fish harvesting, the engineer recommends keeping the fishing rate below 12.5 (thousand fish per unit time). Is this recommendation reasonable, and what population levels y must be sustained to prevent extinction? Explain your reasoning.